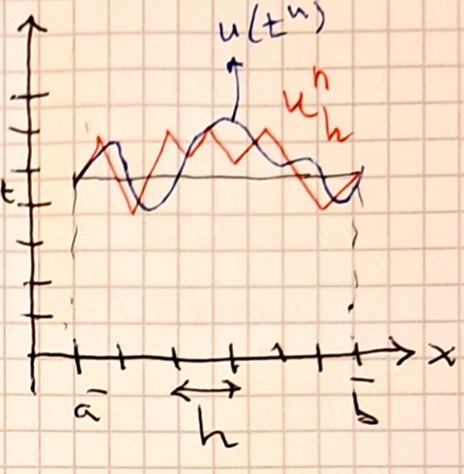
25 NOV 2025 NMPDEs - HPC

1 END of PARABOLIC CHAPTER

CONVERGENCE of Fully-discrete APPX OF PARABOLIC EQS

$$\begin{cases} \dfrac{\partial u}{\partial t} + Lu = f & \text{in } \Omega \times (0,T) \\ + \text{ B.C.s} \\ + \text{ I.C.s} \end{cases}$$



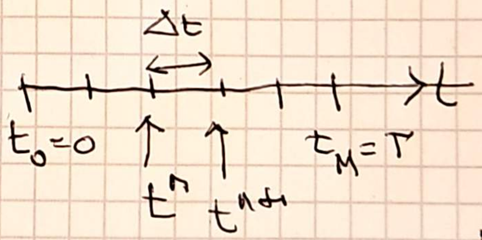
$\downarrow$

Approximated in space using F.E. of degree $r \geq 1$

Approximated in time using

θ-method $\begin{cases} \theta = 1 & \text{BE} \\ \theta = 0 & \text{FE} \\ \theta = 1/2 & \text{CN} \end{cases}$ $\left.\begin{matrix} \\ \\ \end{matrix}\right\}$ order Δt (BE, FE), order Δt^2 (CN)

Convergence result:



$\forall t^n$, $n = 1, \dots, M$

$$\boxed{\| u(t^n) - u_h^n \|_{L^2(\Omega)} \leq C \left[\underbrace{h^r}_{\text{FE in space}} + \underbrace{\Delta t^{q(\theta)}}_{\theta\text{-method in time}} \right]}$$

u_h^n: Fully Discrete Appx Solution

$$q(\theta) = \begin{cases} 1 & \forall \theta \neq 1/2 \\ 2 & \theta = 1/2 \quad \text{CN} \end{cases}$$

TU Graz
Graz University of Technology

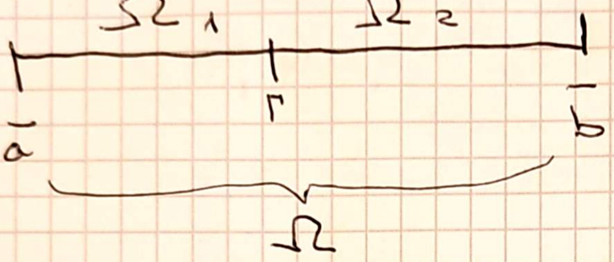
DD - INTRODUCTION

u global solution

$$\text{1D} \quad (P) \begin{cases} Lu = f & \bar{a} < x < \bar{b} \\ u(\bar{a}) = \gamma_1 \\ \mu u'(\bar{b}) = \gamma_2 \end{cases}$$

$$Lu = (-\mu u')' + bu' + \sigma u$$

Ω_i : subdomains
Γ = interface



$$(P) \Leftrightarrow \begin{cases} Lu_1 = f & \text{in } \Omega_1 \\ Lu_2 = f & \text{in } \Omega_2 \end{cases}$$

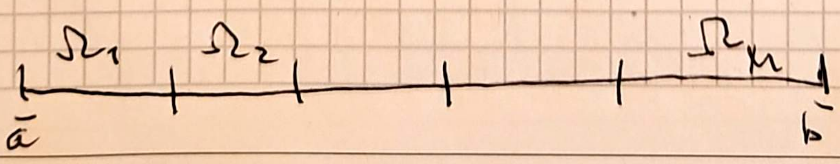
$$(DD\text{-}P) \quad \begin{cases} (D) \rightarrow u_1(\Gamma) = u_2(\Gamma) \\ (N) \rightarrow (\mu u_1')(\Gamma) = (\mu u_2')(\Gamma) \end{cases} \Big\} \text{ interface conditions}$$

$$\begin{cases} u(\bar{a}) = \gamma_1 \\ \mu u'(\bar{b}) = \gamma_2 \end{cases} \Big\} \text{ B.C.}^s$$

then: $u_1 = u|_{\Omega_1}$ $u_2 = u|_{\Omega_2}$ Local solutions

$$(P) \Leftrightarrow (DD\text{-}P)$$

$M \gg 1$

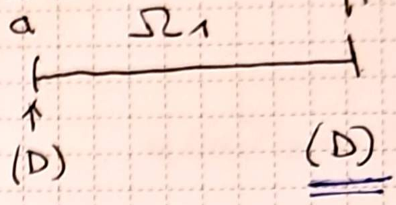


Iterative method: $u_1^{(0)} = a$ on Γ ($\forall a$)

$\forall k \geq 1$

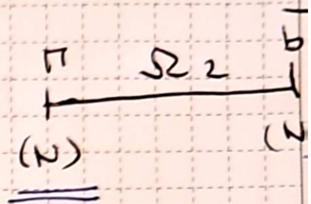
$\theta u_2^{(k-1)} + (1-\theta) u_1^{(k-1)}\big|_\Gamma$

(DN)
$$\begin{cases} L u_1^{(k)} = f & \text{in } \Omega_1 \\ u_1^{(k)} = \boxed{u_2^{(k-1)}} & \text{on } \Gamma \\ + \text{BC} & \text{on } \partial\Omega_1 \cap \partial\Omega \end{cases}$$



$$\begin{cases} L u_2^{(k)} = f & \text{in } \Omega_2 \\ \mu u_2'^{(k)} = \mu u_1'^{(k)} & \text{on } \Gamma \\ + \text{BC}_s & \text{on } \partial\Omega_2 \cap \partial\Omega \end{cases}$$

$\mu u_1'^{(k-1)}$



If $(k) \to \infty$? $u_i^{(k)} \longrightarrow u_i = u|_{\Omega_i}$?

(DN) is called the Dirichlet-Neumann DD method

it is a sequential algorithm / NOT parallel

the red version is PARALLEL (meaning that problems in Ω_1 and Ω_2 can be solved simultaneously).

Blue modification is the DN with Relaxation

$0 < \theta$ is the relaxation parameter

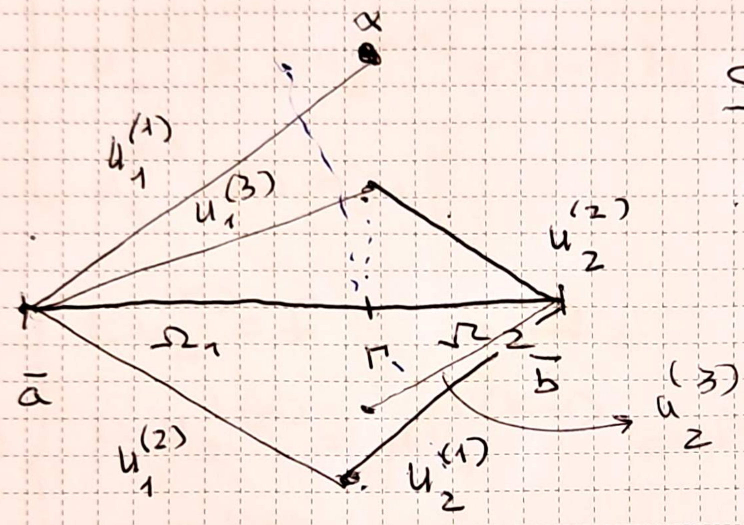
(OUR CHOICE)

Convergence?

example (sequential DN)

$$\begin{cases} Lu := -u'' = \overset{0}{f} & \bar{a} < x < \bar{b} \\ u(\bar{a}) = 0, \; u(\bar{b}) = 0 \end{cases}$$

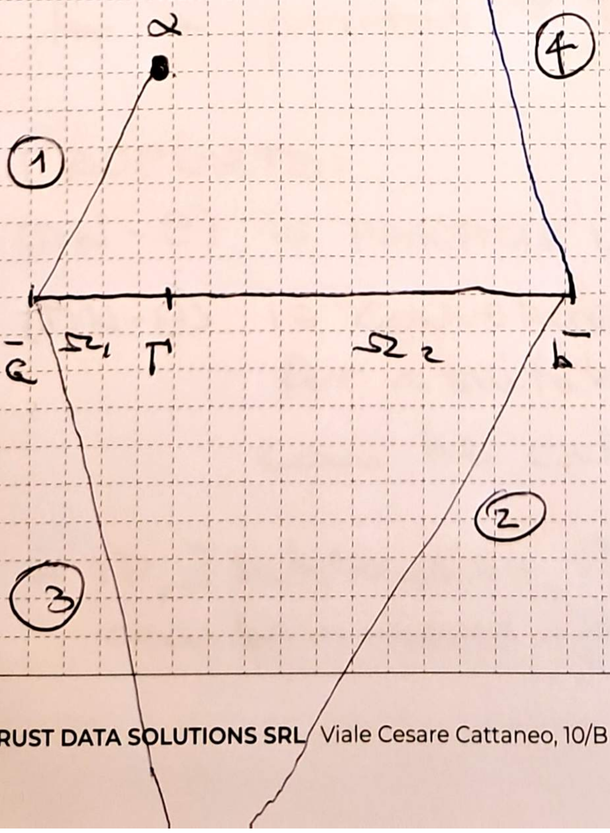
$u(x) \equiv 0 \;\; \forall x$



<u>Convergent!</u>

$u_1^{(k)} \xrightarrow[k \uparrow \infty]{} 0 = u_1$

$u_2^{(k)} \xrightarrow[k \uparrow \infty]{} 0 = u_2$



$\Rightarrow$ $u_1^{(k)} \to \infty$ on Γ

$u_2^{(k)} \to \infty$ on Γ

$\Downarrow$

$\lim u_1^{(k)}$ unbounded

$\lim u_2^{(k)}$ unbounded

$\Rightarrow$ <u>Divergence!</u>

DN with Relaxation: $(DN-\theta)$

$u_1^{(0)} = \alpha$ on Γ $(\forall \alpha)$

$\forall k \geq 1$

$$(D)\begin{cases} L u_1^{(k)} = f & \text{in } \Omega_1 \\ u_1^{(k)} = \theta u_2^{(k-1)} + (1-\theta) u_1^{k-1} & \text{on } \Gamma \\ u_1^{(k)}(a) = \gamma_1 \end{cases}$$

$$(N)\begin{cases} L u_2^{(k)} = f & \text{in } \Omega_2 \\ \mu u_2'^{(k)} = \mu u_1'^{(k-1)} & \text{on } \Gamma \\ \mu u_2'^{(k)}(b) = \gamma_2 \end{cases}$$

for a positive θ to be chosen.

PROPERTY:

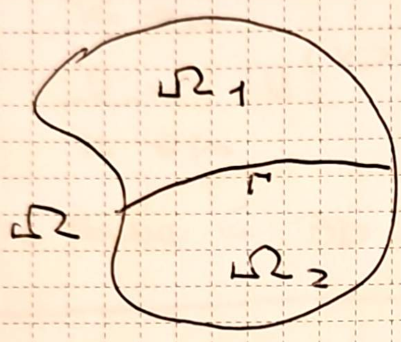
- $(DN-\theta)$ is PARALLEL ☺
- $(DN-\theta)$ is CONVERGENT FOR $0 < \theta < \theta_{max}$ for a suitable θ_{max} that can be calculated.

(in 1D, 2 subdomains $\exists\, \theta_{opt}$ s.t. convergence can be achieved after 1 ITERATION!)

① Generalize to 2D/3D

② Generalize to Many subdomains $\Omega_1 \text{---} \Omega_M$

①



$$(P)\ \begin{cases} Lu = f & \text{in } \Omega \\ + BC_s & \text{on } \partial\Omega \end{cases}$$ global problem.

$\hookrightarrow u(\partial\Omega) = 0$

$\Updownarrow$

$$(DD\text{-}P)\ \begin{cases} Lu_1 = f & \text{in } \Omega_1 \\ u_1(\Gamma) = u_2(\Gamma) & \\ \mu \frac{\partial u_1}{\partial n}(\Gamma) = \mu \frac{\partial u_2}{\partial n}(\Gamma) & \Big\} \text{ interface conditions} \\ Lu_2 = f & \text{in } \Omega_2 \\ u_1(\partial\Omega_1 \cap \partial\Omega) = 0 & \\ u_2(\partial\Omega_2 \cap \partial\Omega) = 0 & \Big\} \text{ B.C.s} \end{cases}$$

$(DN\text{-}\theta)$: $u_1^{(0)}(\Gamma) = \alpha$ given

$\forall k \geq 1$

IN PRACTICE.

FE-approx of (D) $$(D)\ \begin{cases} Lu_1^{(k)} = f & \text{in } \Omega_1 \\ u_1^{(k)} = \theta u_2^{(k-1)} + (1-\theta) u_1^{(k-1)} & \text{on } \Gamma \\ u_1^{(k)} = 0 & \text{on } \partial\Omega_1 \cap \partial\Omega \end{cases}$$

FE-approx of (N) $$(N)\ \begin{cases} Lu_2^{(k)} = f & \text{in } \Omega_2 \\ \mu \frac{\partial u_2^{(k)}}{\partial n} = \mu \frac{\partial u_1^{(k-1)}}{\partial n} & \text{on } \Gamma \\ u_2^{(k)} = 0 & \text{on } \partial\Omega_2 \cap \partial\Omega \end{cases}$$

$\Rightarrow$ PARALLEL $(DN\text{-}\theta)$

$\exists \vartheta_{max}$! if $0 < \theta < \vartheta_{max}$ then (DN - θ) converges $\Leftrightarrow$

$$\lim_{k \to \infty} u_1^{(k)} = u_1 \; (= u|_{\Omega_1})$$ local solutions

$$\lim_{k \to \infty} u_2^{(k)} = u_2 \; (= u|_{\Omega_2})$$

The same in 3D.

Property: IN the FE case, the RATE of CONVERGENCE is INDEPENDENT of h $\Rightarrow$ OPTIMALITY.

When using many subdomains $M \gg 1$

is the method still OPTIMAL?

is it SCALABLE?

In general the answer is no! We will see it

TRUST DATA SOLUTIONS

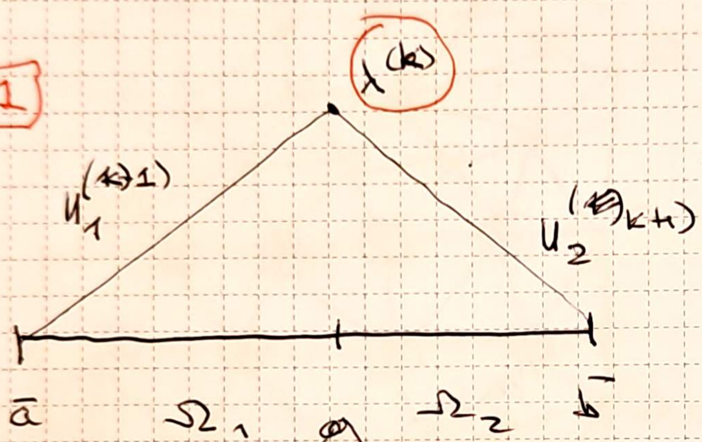
DD-7

~~FE (DN-B)~~

NN ALGORITHM

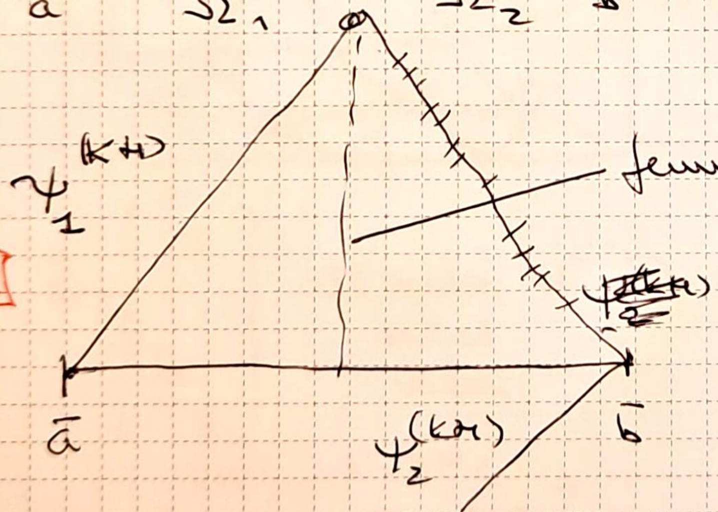
1D $\begin{cases} Lu = -u'' = 0 & \bar{a} < x < \bar{b} \\ u(\bar{a}) = 0 \quad u(\bar{b}) = 0 \end{cases}$

$\boxed{1}$



2 D problems

$\boxed{2}$



jump of the 2 derivatives

$u_1'^{(1)} - u_2'^{(2)}$

2 N problems

$\boxed{3}$ $\lambda^{(k+1)} = \lambda^{(k)} - \theta \left(\sigma_1 \psi_{1|\Gamma}^{(k+1)} - \sigma_2 \psi_{2|\Gamma}^{(k+1)} \right)$

jump of values

σ_1, σ_2 2 positive coefficients

$\Rightarrow$ PROPERTY :

$\lambda^{(k)} \xrightarrow[k \to \infty]{} u_1(\Gamma) = u_2(\Gamma) = u(\Gamma)$

$\Rightarrow$ convergence if $0 < \theta < \theta_{max}$

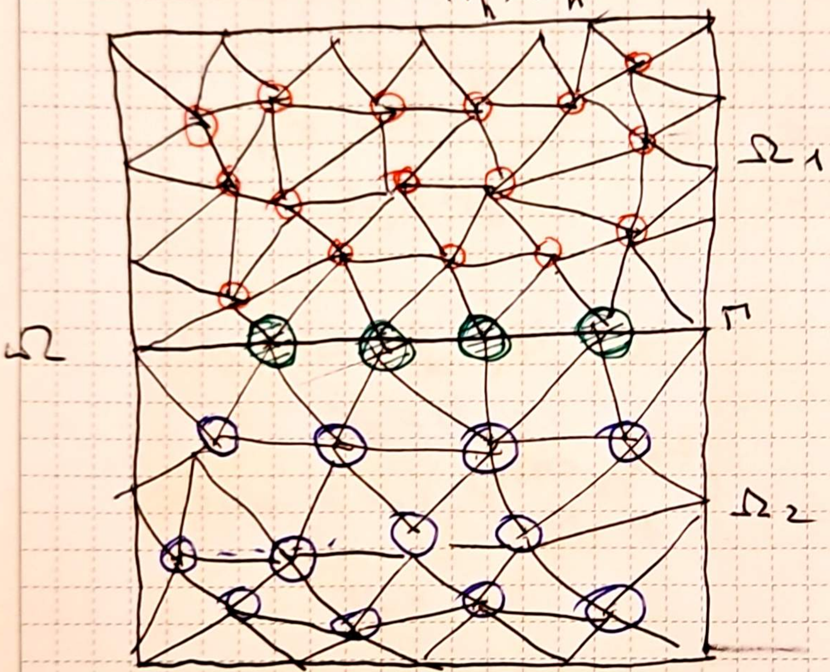
TRUST
DATA SOLUTIONS

DD-8

$$\begin{cases} Lu = f & \text{in } \Omega \\ + \text{B.Cs} & \text{on } \partial\Omega \end{cases} \quad (\text{Dirichlet})$$

FE Appx $\longrightarrow$ $\underbrace{\boxed{A \vec{u} = \vec{f}}}_{N_h \times N_h}$ linear algebraic system.

$r = 1$ linear FEs



N_1 ○ interior FE nodes in Ω_1

N_Γ ○ interface FE nodes on Γ

N_2 ○ interior nodes in Ω_2

$N_1 = 16$

$N_\Gamma = 4$ $\quad N_h = N_1 + N_\Gamma + N_2$

$N_2 = 12$ $\quad = 32$

$\vec{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_{N_h} \end{bmatrix}$ all ~~nodes~~ of Ω nodal values

$\vec{u} = \begin{bmatrix} \vec{u}_1 \\ \vec{u}_2 \\ \vec{\lambda} \end{bmatrix}$ ○ ○ ○ $\quad \vec{\lambda} = \vec{u}\big|_\Gamma$ nodal values on Γ

$$A \overset{?}{=} \underbrace{\begin{bmatrix} \overset{N_1 \times N_1}{A_{11}} & A_{12} & A_{1\Gamma} \\ A_{21} & \boxed{A_{22}}^{\,N_2 \times N_2} & A_{2\Gamma} \\ A_{\Gamma 1} & A_{\Gamma 2} & \boxed{A_{\Gamma\Gamma}}_{\,N_\Gamma \times N_\Gamma} \end{bmatrix}}_{N_h} \begin{bmatrix} \vec{u}_1 \\ \vec{u}_2 \\ \vec{\lambda} \end{bmatrix} = \begin{bmatrix} \vec{f}_1 \\ \vec{f}_2 \\ \vec{f}_\Gamma \end{bmatrix} \quad (*)$$

DD - 9

TRUST
DATA SOLUTIONS

Meaning of submatrices A_{ij}:

$$A = (a_{ij}) = a(\varphi_i, \varphi_j) = \int_\Omega \dots$$

global stiffness matrix

$$A_{11} = a(\varphi_i^{(1)}, \varphi_j^{(1)}) \quad \text{basis functions associated with nodes in } \Omega_1$$

$N_1 \times N_1$

$$A_{22} = a(\varphi_k^{(2)}, \varphi_m^{(2)}) \quad " \quad \Omega_2$$

$N_2 \times N_2$

$$A_{\Gamma\Gamma} = a(\varphi_p^{(\Gamma)}, \varphi_q^{(\Gamma)}) \quad " \quad \Gamma$$

$N_\Gamma \times N_\Gamma$

$A_{ij} \quad i \neq j \Leftrightarrow$ coupling between basis functions belonging to different "subdomains"

$\Rightarrow A_{12} = 0 \quad A_{21} = 0$

$$\Rightarrow \begin{bmatrix} A_{11} & 0 & A_{1\Gamma} \\ 0 & A_{22} & A_{2\Gamma} \\ A_{\Gamma 1} & A_{\Gamma 2} & A_{\Gamma\Gamma} \end{bmatrix} \cdot \begin{bmatrix} \vec{u}_1 \\ \vec{u}_2 \\ \vec{\lambda} \end{bmatrix} = \begin{bmatrix} \vec{f}_1 \\ \vec{f}_2 \\ \vec{f}_\Gamma \end{bmatrix}$$

TRUST DATA SOLUTIONS

Let us use Gaussian elimination to "eliminate" $\vec{u}_1$ and $\vec{u}_2$! (this is a forward step)

$$A_{11}\vec{u}_1 + A_{1\Gamma}\vec{\lambda} = \vec{f}_1 \rightarrow \vec{u}_1 = A_{11}^{-1}(\vec{f}_1 - A_{1\Gamma}\vec{\lambda})$$

$$A_{22}\vec{u}_2 + A_{2\Gamma}\vec{\lambda} = \vec{f}_2 \rightarrow \vec{u}_2 = A_{22}^{-1}(\vec{f}_2 - A_{2\Gamma}\vec{\lambda})$$

3rd eq:

$$A_{\Gamma 1}\vec{u}_1 + A_{\Gamma 2}\vec{u}_2 + A_{\Gamma\Gamma}\vec{\lambda} = \vec{f}_\Gamma$$

$$\Rightarrow \quad A_{\Gamma 1}A_{11}^{-1}(\vec{f}_1 - A_{1\Gamma}\vec{\lambda}) + A_{\Gamma 2}A_{22}^{-1}(\vec{f}_2 - A_{2\Gamma}\vec{\lambda}) + A_{\Gamma\Gamma}\vec{\lambda} = \vec{f}_\Gamma$$

$$\Rightarrow \Big[\overbrace{\big(-A_{\Gamma 1}A_{11}^{-1}A_{1\Gamma} - A_{\Gamma 2}A_{22}^{-1}A_{2\Gamma}\big) + A_{\Gamma\Gamma}}^{N_\Gamma \times N_\Gamma \text{ matrix } (\Sigma)}\Big]\overbrace{\vec{\lambda}}^{\text{nodal values on } \Gamma\ (N_\Gamma)} =$$

$$\vec{f}_\Gamma - A_{\Gamma 1}A_{11}^{-1}\vec{f}_1 - A_{\Gamma 2}A_{22}^{-1}\vec{f}_2 \equiv \vec{X}_\Gamma \quad \text{known vector } (N_\Gamma)$$

$$\Leftrightarrow \quad \boxed{\Sigma\vec{\lambda} = \vec{X}_\Gamma}$$

Schur complement system of $(*)$

Observe $\quad A_{\Gamma\Gamma} = a(\varphi_p^{(\Gamma)}, \varphi_q^{(\Gamma)}) = \underbrace{\int_{\Omega_1} \cdots}_{A_{\Gamma\Gamma}^{(1)}} + \underbrace{\int_{\Omega_2} \cdots}_{A_{\Gamma\Gamma}^{(2)}}$

$$\left[\underbrace{\left(A^{(1)}_{\Gamma\Gamma} - A_{\Gamma 1} A_{11}^{-1} A_{1\Gamma}\right)}_{\Sigma^{(1)}} + \underbrace{\left(A^{(2)}_{\Gamma\Gamma} - A_{\Gamma 2} A_{22}^{-1} A_{2\Gamma}\right)}_{\Sigma^{(2)}}\right] \vec{\lambda} = \vec{X}_\Gamma$$

$$\Sigma^{(1)} + \Sigma^{(2)} = \Sigma$$

$\Rightarrow$ $\Sigma^{(i)}$ <u>local</u> Schur complements in Ω_i

Σ global Schur complement.

FONDAZIONE
C.I.M.E.
CENTRO INTERNAZIONALE MATEMATICO ESTIVO
INTERNATIONAL MATHEMATICAL SUMMER CENTER

C/O DIPARTIMENTO DI MATEMATICA E INFORMATICA "U.DINI"
VIALE MORGAGNI 67/A, 50134 - FIRENZE
TEL / FAX +39 055 2751427

e-mail: cime@math.unifi.it
web site: www.math.unifi.it/cime